

PURE MATHEMATICS WITH MECHANICS • UPPER SIXTH

UPPER SIXTH — PURE MATHEMATICS WITH MECHANICS

CHAPTER 16

NEWTON'S LAWS OF MOTION

SUGGESTED TIME: 5 PERIODS

IN THIS CHAPTER

This chapter introduces Newton's three laws of motion, the cornerstone of classical mechanics. Students will learn to analyse forces, predict accelerations, and understand the limits of these laws in modern physics.

- State and explain Newton's first law (law of inertia) and identify inertial reference frames.
- Apply Newton's second law ($F = ma$) to solve problems involving constant and variable forces.
- Use the concept of momentum and impulse to analyse collisions and impacts.
- State Newton's third law and correctly draw action-reaction pairs in free-body diagrams.
- Construct and interpret free-body diagrams for particles and rigid bodies in equilibrium and motion.
- Solve real-life problems (e.g., vehicle braking, rocket thrust, tension in ropes) using Newton's laws.



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CHAPTER OBJECTIVES

- State and explain Newton's first law (law of inertia) and identify inertial reference frames.
- Apply Newton's second law ($F = ma$) to solve problems involving constant and variable forces.
- Use the concept of momentum and impulse to analyse collisions and impacts.
- State Newton's third law and correctly draw action–reaction pairs in free–body diagrams.
- Construct and interpret free–body diagrams for particles and rigid bodies in equilibrium and motion.
- Solve real–life problems (e.g., vehicle braking, rocket thrust, tension in ropes) using Newton's laws.
- Discuss the limitations of Newtonian mechanics (high speeds, quantum scale) and outline its extensions.
- Communicate mechanical reasoning clearly with appropriate mathematical notation and diagrams.

PREREQUISITES

- Vectors: addition, resolution, and scalar product (Lower Sixth).
- Kinematics: displacement, velocity, acceleration, and equations of uniform acceleration.

- Concept of force as a push or pull and its measurement in newtons.
- Basic algebra and calculus (differentiation of simple functions).
- Understanding of inertial and non-inertial frames from introductory physics.

1 INERTIA AND THE FIRST LAW

Imagine a loaded taxi speeding down the steep hills of Buea; when the driver brakes suddenly, the passengers lurch forward. Why do they keep moving? This everyday scene hides the fundamental principles that govern all motion, from a football kick in Yaoundé to the launch of a satellite. The taxi slows because the brakes act on it; but nothing acts directly on the passengers, so they continue with the velocity they had. This simple observation, which every Cameroonian has felt in a bendskin or a coaster bus, took humanity more than two thousand years to understand properly. In this section we trace that intellectual journey, define **inertia** precisely, state **Newton's First Law**, and learn to recognise the frames of reference in which it holds.

1.1 Historical background: Aristotle, Galileo, Newton

Aristotle's view (4th century BC). Aristotle taught that the “natural state” of a body is rest, and that a force is needed to maintain any motion. Push a chair and it moves; stop pushing and it stops. This agrees with naive everyday experience, and for nearly twenty centuries it was accepted as truth.

Galileo's breakthrough (early 17th century). Galileo Galilei asked a subtler question: why does the chair stop? His answer: because of friction, not because motion “naturally dies”. He imagined (and performed) experiments with balls rolling down one inclined plane and up another. The ball always rises to almost the same height it started from; and as the second plane is made less and less steep, the ball travels farther and farther to regain that height. In the limit of a horizontal, perfectly smooth plane, the ball would never regain its height — it would roll on forever at constant speed. Galileo concluded:

In the absence of friction, a body in motion on a horizontal surface continues moving uniformly, with no force required to sustain the motion.

Force is therefore needed to change motion, not to maintain it. This was a revolution in thinking: the “natural state” includes uniform motion, not only rest.

Newton's synthesis (1687). Isaac Newton, in his *Philosophiæ Naturalis Principia Mathematica* (1687), elevated Galileo's insight to the status of a fundamental law — his First Law of Motion — and built upon it the whole edifice of classical mechanics. Rest and uniform rectilinear motion are dynamically equivalent: neither requires a cause; only changes of velocity do.

DID YOU KNOW?

A thought experiment that changed the world

Galileo's inclined-plane argument is one of the first great “thought experiments” in science: he could not eliminate friction in his laboratory, so he eliminated it in his mind, and reasoned about the limiting case. This method — idealise, then reason rigorously — remains at the heart of theoretical physics. Newton later wrote that he had “stood on the shoulders of giants”; Galileo was the first of them.

1.2 Inertia and mass

The taxi passengers lurch forward because their bodies “resist” the change of velocity imposed by the braking vehicle. This resistance to any change in the state of motion is called **inertia**.

DEFINITION Inertia

Inertia is the property of every material body by which it resists any change in its velocity — whether a change of speed, of direction, or the start of motion from rest.

Experience shows that some bodies have more inertia than others. It is far easier to stop a rolling football than a rolling water drum moving at the same speed; easier to push–start a bicycle than a lorry. The physical quantity that measures inertia is the **mass**.

DEFINITION Mass as a measure of inertia

The **mass** m of a body is the quantitative measure of its inertia: the greater the mass, the more the body resists changes in its velocity. Mass is a scalar quantity, measured in kilograms (kg) in the SI system.

WARNING Mass is not weight

Mass (inertia) must not be confused with weight, which is the gravitational force exerted on the body. A bag of cement of mass 50 kg has the same inertia on Earth and on the Moon, but its weight is about six times smaller on the Moon (since $g_{\text{Moon}} \approx 1.6 \text{ m s}^{-2}$ against $g_{\text{Earth}} \approx 9.81 \text{ m s}^{-2}$). Inertia is intrinsic; weight depends on the local gravity. In the GCE examination, never write “the mass of the bag is 50 N” — newtons measure force, not mass.

1.3 Statement of Newton’s First Law

PROPERTY Newton’s First Law (Law of Inertia)

A body remains at rest, or continues to move in a straight line with constant velocity, unless it is compelled to change that state by a net external force acting upon it. Equivalently:

$$\vec{F}_{\text{net}} = \sum \vec{F} = \vec{0} \quad \Longleftrightarrow \quad \vec{v} = \text{constant (possibly } \vec{0}).$$

Several remarks are essential for a rigorous understanding:

- The law concerns the net (resultant) force. A body may be acted upon by several forces; if they balance, the velocity does not change.

- “Constant velocity” means constant vector velocity: constant speed and constant direction. A car going round a roundabout at steady speed is not covered by the First Law — its direction changes, so a net force must act on it (we shall see in the chapter on circular motion that this force points towards the centre).
- Rest is simply the special case $\vec{v} = \vec{0}$. The First Law thus unifies “bodies at rest stay at rest” and “bodies in uniform motion stay in uniform motion”.
- The law is an equivalence: zero net force implies constant velocity, and conversely, observed constant velocity implies zero net force. Both directions are used in problems.

KEY POINTS What the First Law really says

- No net force $\Rightarrow$ no acceleration: $\vec{a} = \vec{0}$.
- Force is the cause of change of velocity, not of velocity itself.
- A body in uniform rectilinear motion is dynamically indistinguishable from a body at rest.

1.4 Inertial reference frames

Here lies a subtlety that the GCE examiners appreciate. Is the First Law true in every frame of reference? Consider again the braking taxi. Relative to the road, the passengers simply continue at constant velocity — the First Law holds. But relative to the taxi (a decelerating frame), the passengers suddenly accelerate forward with no visible force pushing them. The First Law appears violated. The taxi, while braking, is not a suitable frame for Newton’s mechanics.

DEFINITION Inertial frame of reference

An **inertial frame** (or Galilean frame) is a frame of reference in which a body subject to no net force moves with constant velocity (in particular, remains at rest if initially at rest). Newton’s laws are valid only in inertial frames.

The Earth is, strictly speaking, not inertial: it rotates on its axis and revolves around the Sun, so it is permanently accelerating. However, for the durations and distances of ordinary laboratory experiments (and of GCE problems!), these accelerations are so small that **a frame fixed to the ground may be treated as inertial** to an excellent approximation. Moreover, any frame moving at constant velocity relative to an inertial frame is itself inertial: a train travelling smoothly at constant speed on straight rails is just as good a frame as the platform.

METHOD Testing whether a frame is inertial

1. Free-body test. Isolate a body from all interactions (or arrange balanced forces). Ob-

serve its motion relative to the frame. If it stays at rest or moves uniformly in a straight line, the frame passes the test.

2. Foucault pendulum test. Suspend a long pendulum and set it swinging. The plane of oscillation is fixed relative to inertial space. If the plane appears to rotate relative to your frame (as Foucault demonstrated publicly in Paris in 1851), your frame is rotating — hence non-inertial. On Earth the plane rotates slowly, revealing that the Earth frame is only approximately inertial.
3. Plumb-line test. Hang a plumb line in the frame. If it hangs vertically and steadily while no horizontal force acts on the bob, the frame has no horizontal acceleration; if it tilts or swings without cause, the frame is accelerating.

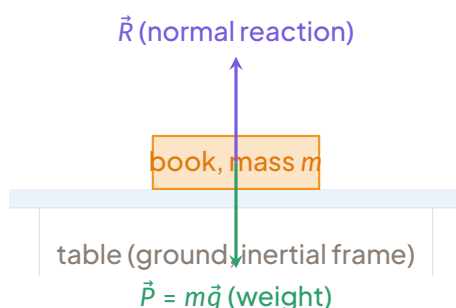
1.5 Identifying forces on a body at rest or in uniform motion

The First Law gives us a powerful diagnostic tool: **if a body is at rest or in uniform rectilinear motion (in an inertial frame), then the vector sum of the forces on it is zero.** We use this to identify and relate forces. The systematic procedure is: isolate the body, list only the forces acting on it, draw the free-body diagram, then write $\sum \vec{F} = \vec{0}$.

Example 1: the book on the table. A book lies motionless on a horizontal table. Two forces act on it:

- its **weight** $\vec{P} = m\vec{g}$, directed vertically downward (the Earth's pull);
- the **normal reaction** $\vec{R}$ of the table, directed vertically upward, perpendicular to the contact surface (the table's support).

Since the book is at rest, the First Law gives $\vec{P} + \vec{R} = \vec{0}$, hence $R = P = mg$.

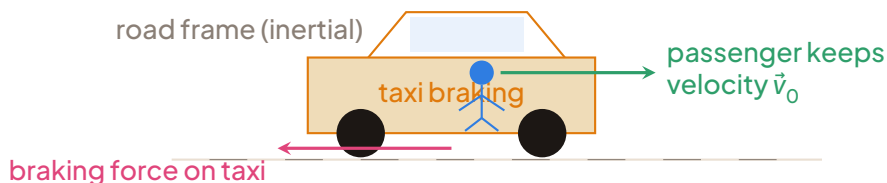


Free-body diagram of a book at rest on a horizontal table: the two forces balance, so the net force is zero and the book remains at rest (First Law).

Example 2: the hockey puck on ice. A puck struck on very smooth ice slides in a straight line at nearly constant speed long after the stick has left it. Horizontally, almost no force acts (friction is tiny); vertically, weight and the ice's reaction cancel. The net force is nearly zero, so the motion is nearly uniform — a beautiful, direct illustration of the First Law, and of why Galileo insisted

on eliminating friction in his reasoning.

Example 3: the passenger in the braking taxi. Before braking, the passenger moves with the taxi at velocity $\vec{v}_0$; the forces on him (weight, seat reaction, friction with the seat) are balanced. When the taxi brakes, the taxi experiences a backward force, but the passenger does not (at first). By the First Law, he continues at $\vec{v}_0$ relative to the road — which looks, from inside the taxi, like a forward lurch. The seat belt then provides the real backward force that decelerates him with the vehicle.



When the taxi brakes, a backward force acts on the taxi but not directly on the passenger. By inertia, the passenger continues forward at the initial velocity $\vec{v}_0$ until the seat belt or seat exerts a retarding force.

WARNING “No force” versus “balanced forces”

A classic trap. The First Law is satisfied in both situations — a body deep in space with no force at all, and the book on the table with balanced forces — because in both cases the net force is zero and the acceleration is zero. But physically the situations differ: only the truly force-free body can serve to define an inertial frame. A frame in which a genuinely force-free body accelerates is non-inertial, even if other bodies with balanced forces happen to move uniformly in it. In examinations, always say “the resultant (net) force is zero”, never “there is no force”, when forces are present but balanced.

WORKED EXAMPLE 1 Forces on a crate in a moving lorry

A crate of mass $m = 40 \text{ kg}$ rests on the flat floor of a lorry travelling along a straight horizontal road at a constant speed of 60 km h^{-1} . The crate does not slide.

1. List the forces acting on the crate and give the value of the resultant force.
2. The driver then brakes. Explain, using the First Law, what happens to the crate if the floor is very smooth.

SOLUTION

Solution.

1. The lorry moves uniformly in a straight line, so it constitutes (to a very good approximation) an inertial frame, and the crate is at rest in it. The forces on the crate are: its

weight $\vec{P} = m\vec{g}$ (vertically downward) and the floor's normal reaction $\vec{R}$ (vertically upward). Since the velocity is constant, the First Law gives

$$\vec{P} + \vec{R} = \vec{0} \implies R = P = mg = 40 \times 9.81 = 392.4 \text{ N},$$

and the resultant force is $\vec{F}_{\text{net}} = \vec{0}$. Note carefully: no horizontal force is needed, because uniform motion requires none. (If the floor were rough, a tiny static friction could exist, but it is not required and we take it as zero here.)

- When the lorry brakes, the lorry slows but no horizontal force acts on the crate (smooth floor, negligible friction). By the First Law the crate keeps its velocity of 60 km h^{-1} relative to the road: it slides forward relative to the lorry, towards the cabin, until it meets an obstacle that exerts a retarding force on it. This is exactly why loads in lorries must be strapped down.

EXERCISE 1 Thinking with the First Law

- A ball of mass 0.45 kg rests on the pitch at the Ahmadou Ahidjo stadium. Draw its free-body diagram and compute the reaction of the ground (take $g = 9.81 \text{ m s}^{-2}$).
- A parachutist falls at constant vertical speed. Is the resultant force on him zero? Name the forces involved.
- Passengers in a coaster bus are thrown to the right when the bus turns. In which direction did the bus turn? Justify with the First Law.
- A frame of reference is fixed in a lift. In which of the following cases is this frame (approximately) inertial: (a) the lift is at rest; (b) the lift rises at constant speed; (c) the lift accelerates upward? Justify each answer.

SOLUTION Exercise 1

- Two forces act on the ball: its weight $\vec{P} = m\vec{g}$ downward and the ground's normal reaction $\vec{R}$ upward. The ball is at rest, so by the First Law $\vec{P} + \vec{R} = \vec{0}$, hence

$$R = P = mg = 0.45 \times 9.81 \approx 4.41 \text{ N}, \quad \text{directed vertically upward.}$$
- Yes. Constant velocity (uniform rectilinear motion) implies, by the First Law, that the net force is zero. The forces are his weight $\vec{P}$ downward and the air resistance (drag) $\vec{f}$ upward, with $f = P$: the two forces balance. (This constant falling speed is called the terminal velocity.)
- The passengers' bodies tend to continue in a straight line (inertia) while the bus turns

beneath them. If they are thrown to the right, the bus turned to the left: relative to the bus, their rectilinear motion carries them towards the outside of the bend.

4. (a) At rest: inertial — a frame at rest relative to the ground is inertial to the usual approximation. (b) Rising at constant speed: inertial — it moves with constant velocity relative to an inertial frame. (c) Accelerating upward: not inertial — a body released inside would appear to accelerate downward relative to the lift with no corresponding real force, violating the First Law.

KEY POINTS Key points of this section

- Aristotle: force maintains motion (wrong). Galileo: friction stops motion; without it, uniform motion persists. Newton: this becomes the First Law.
- **Inertia** is the resistance of a body to changes of velocity; **mass** measures inertia (scalar, in kg); it must not be confused with weight (a force, in N).
- First Law: $\sum \vec{F} = \vec{0} \iff \vec{v} = \text{constant}$. Force changes velocity; it does not maintain it.
- Newton's laws hold only in **inertial frames**; the ground frame is inertial to an excellent approximation for everyday motions, and any frame in uniform rectilinear motion relative to it is also inertial.
- A body at rest or in uniform motion has balanced forces (net force zero) — which is not the same as having no forces at all.

2 FORCE, ACCELERATION AND THE SECOND LAW

In the previous section we met Newton's First Law: a body keeps its state of rest or of uniform motion in a straight line unless a net external force compels it to change. That law tells us when motion changes; it does not tell us by how much. If you push a wheelbarrow full of sand at the market in Bamenda and then push the same wheelbarrow when it is empty, you feel immediately that the same push produces very different changes of motion. Newton's Second Law makes this observation precise: it is the quantitative heart of dynamics, linking **force**, **mass** and **acceleration** in one compact equation.

2.1 From intuition to the law: momentum

Experience suggests two facts. First, the harder you push (the greater the force), the faster the velocity changes. Second, the heavier the object (the greater the mass), the harder it is to change its velocity. Newton captured both ideas by introducing a new quantity.

DEFINITION Momentum

The **momentum** of a body of mass m moving with velocity $\vec{v}$ is the vector quantity

$$\vec{p} = m\vec{v}.$$

Its SI unit is the kilogram metre per second, kg m s^{-1} (equivalently the newton second, N s). Momentum points in the same direction as the velocity.

Momentum measures the “quantity of motion” of a body. A 1200 kg car moving at 10 m s^{-1} has the same magnitude of momentum as a 60 kg athlete sprinting at 20 m s^{-1} — both are equally “hard to stop” in this precise sense.

PROPERTY Newton’s Second Law — momentum form

The **rate of change of momentum** of a body equals the **net external force** acting on it, and takes place in the direction of that force:

$$\sum \vec{F} = \frac{d\vec{p}}{dt}.$$

(This statement, and its derivation of $\sum \vec{F} = m\vec{a}$ for constant mass, are examinable.)

This is the form Newton actually gave, and it is the most general one: it applies even when the mass of the system changes.

2.2 Constant mass: the familiar equation $\sum \vec{F} = m\vec{a}$

For a body whose mass does not change, the derivation is immediate:

$$\sum \vec{F} = \frac{d\vec{p}}{dt} = \frac{d}{dt}(m\vec{v}) = m \frac{d\vec{v}}{dt} = m\vec{a}.$$

PROPERTY Second Law for constant mass

If the mass m of the system is constant, then

$$\sum \vec{F} = m\vec{a},$$

where $\sum \vec{F}$ is the vector sum of all external forces and $\vec{a}$ is the acceleration of the centre of mass. For variable mass systems one must use the general form $\sum \vec{F} = \frac{d(m\vec{v})}{dt}$.

Several consequences deserve emphasis:

- The equation is **vectorial**: it is really three scalar equations, one per axis. In two dimensions,

$$\sum F_x = ma_x \text{ and } \sum F_y = ma_y.$$

- The SI unit of force follows from the law: $1 \text{ N} = 1 \text{ kg m s}^{-2}$. A force of 1 N gives a mass of 1 kg an acceleration of 1 m s^{-2} .
- Weight is just the Second Law applied to gravity: $\vec{W} = m\vec{g}$, so $W = mg$ with $g \approx 9.81 \text{ m s}^{-2}$.
- If $\sum \vec{F} = \vec{0}$ then $\vec{a} = \vec{0}$: the First Law is recovered as a special case.

WARNING Vectors before numbers

Do not mix scalar and vector quantities. $\sum \vec{F} = m\vec{a}$ is a vector equation. Always resolve forces into components along chosen axes before substituting numbers. Writing “ $F = ma$ ” with magnitudes of forces that are not parallel to the motion is the most common source of error at GCE. Also remember: $m\vec{a}$ is not an extra force to be drawn on the diagram — it is the result of the real forces.

2.3 Impulse and change of momentum

Integrating the Second Law with respect to time gives a result of great practical value, especially for collisions and blows, where forces act briefly but intensely.

DEFINITION Impulse

The **impulse** of a force $\vec{F}$ acting between times t_1 and t_2 is

$$\vec{J} = \int_{t_1}^{t_2} \vec{F} dt.$$

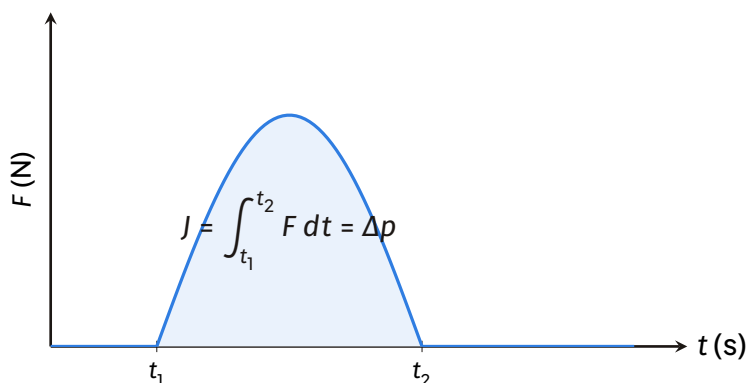
Its unit is the newton second, N s .

PROPERTY Impulse–momentum theorem

The impulse of the net force equals the change of momentum of the body:

$$\vec{J} = \int_{t_1}^{t_2} \sum \vec{F} dt = \Delta \vec{p} = m\vec{v}_2 - m\vec{v}_1 \quad (\text{constant mass}).$$

The proof is one line: since $\sum \vec{F} = d\vec{p}/dt$, integrating both sides from t_1 to t_2 gives $\int \sum \vec{F} dt = \vec{p}(t_2) - \vec{p}(t_1)$. Geometrically, the impulse is the area under the force–time graph.



The impulse of a force is the area under the F - t graph; it equals the change of momentum.

This explains everyday experience: a fielder catching a fast cricket ball draws his hands back, lengthening the contact time Δt ; since $J = \Delta p$ is fixed by the ball, a longer Δt means a smaller average force on the hands.

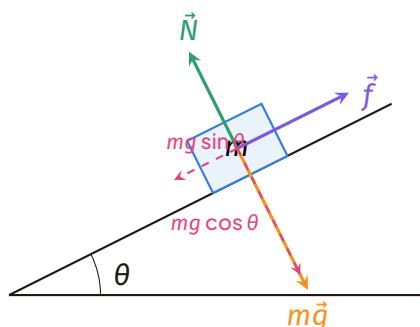
2.4 A systematic method for dynamics problems

METHOD Solving a dynamics problem

1. Choose the system: decide which body (or bodies) you analyse, and isolate it mentally from its surroundings.
2. Draw the free-body diagram: represent all external forces on the system (weight, normal reaction, tension, friction, applied forces) — nothing else.
3. Apply $\sum \vec{F} = m\vec{a}$ in components: choose axes wisely (often parallel and perpendicular to the motion), resolve every force, and write one scalar equation per axis.
4. Solve the kinematics: once $\vec{a}$ is known, use the constant-acceleration formulae ($v = u + at$, $s = ut + \frac{1}{2}at^2$, $v^2 = u^2 + 2as$) to answer the question.

2.5 Worked model 1: block on an inclined plane

Consider a block of mass m on a rough plane inclined at angle θ to the horizontal, with coefficient of kinetic friction μ_k . The forces are the weight $m\vec{g}$, the normal reaction $\vec{N}$, and friction $\vec{f}$ opposing the motion.



Free-body diagram on an inclined plane: the weight is resolved into components $mg \sin \theta$ (down the slope) and $mg \cos \theta$ (into the plane).

Take axes x down the slope and y perpendicular to it. Resolving:

$$\begin{aligned} y : N - mg \cos \theta &= 0 \quad \Rightarrow \quad N = mg \cos \theta, \\ x : mg \sin \theta - f &= ma, \quad f = \mu_k N = \mu_k mg \cos \theta. \end{aligned}$$

Hence

$$a = g (\sin \theta - \mu_k \cos \theta).$$

Note the cancellation of m : all blocks slide down a given rough plane with the same acceleration, a fact Galileo would have appreciated.

2.6 Worked model 2: the Atwood machine

Two masses $m_1 > m_2$ hang from a light inextensible string passing over a smooth pulley. The string tension T is the same on both sides; both masses share the same magnitude of acceleration a (the string is inextensible). Applying the Second Law to each mass separately, taking the direction of motion as positive:

$$\begin{aligned} m_1 g - T &= m_1 a, \\ T - m_2 g &= m_2 a. \end{aligned}$$

Adding eliminates T :

$$a = \frac{(m_1 - m_2)}{m_1 + m_2} g, \quad T = \frac{2m_1 m_2}{m_1 + m_2} g.$$

WARNING One system or two?

A frequent mistake is to write $m_1 g - m_2 g = m_1 a$ (forgetting that m_2 also accelerates). When two bodies move together, either treat each body separately with its own equation, or treat the whole system with total mass $m_1 + m_2$ and external driving force $(m_1 - m_2)g$. Never mix the two viewpoints in a single equation.

2.7 Worked model 3: braking distance of a car

A car of mass m travelling at speed u brakes with a constant total resistive force F (braking force plus friction). The Second Law gives the deceleration $a = F/m$. Using $v^2 = u^2 + 2as$ with $v = 0$:

$$0 = u^2 - 2\frac{F}{m}s \quad \Rightarrow \quad s = \frac{mu^2}{2F}.$$

The stopping distance grows with the square of the speed: doubling your speed quadruples the braking distance. This is why speed limits near schools in Yaoundé or Douala are a matter of physics, not bureaucracy.

WORKED EXAMPLE 2 Impulse of a football kick

A football of mass 0.43 kg, initially at rest, is kicked and leaves the foot at 24 m s^{-1} . The foot is in contact with the ball for 0.012 s. Find (a) the impulse delivered, (b) the average force exerted on the ball.

SOLUTION

Solution. (a) By the impulse–momentum theorem, with the ball initially at rest:

$$J = \Delta p = mv - 0 = 0.43 \times 24 = 10.32 \text{ N s}.$$

(b) Assuming a constant average force over the contact time,

$$F_{\text{avg}} = \frac{J}{\Delta t} = \frac{10.32}{0.012} = 860 \text{ N}.$$

That is roughly the weight of an 88 kg person, applied for a hundredth of a second — enough to explain why a poorly struck ball stings the foot.

WORKED EXAMPLE 3 Block pulled on a rough horizontal table

A crate of mass 25 kg is pulled along a horizontal floor by a horizontal force of 120 N. The coefficient of kinetic friction is $\mu_k = 0.30$. Take $g = 9.81 \text{ m s}^{-2}$. Find the acceleration of the crate and the distance covered in 4.0 s starting from rest.

SOLUTION

Solution. Free-body diagram: weight mg down, normal reaction N up, applied force $F = 120 \text{ N}$, friction f opposing motion.

Vertically: $N = mg = 25 \times 9.81 = 245.25 \text{ N}$.

Friction: $f = \mu_k N = 0.30 \times 245.25 = 73.6 \text{ N}$.

Horizontally (Second Law):

$$F - f = ma \quad \Rightarrow \quad a = \frac{120 - 73.6}{25} = 1.86 \text{ m s}^{-2}.$$

Kinematics from rest:

$$s = \frac{1}{2}at^2 = \frac{1}{2} \times 1.86 \times (4.0)^2 = 14.9 \text{ m.}$$

2.8 A glimpse of variable mass: rocket propulsion

So far the mass of the system was constant. A rocket is different: it ejects burnt fuel backwards, so its mass decreases as it accelerates. The simple formula $\sum \vec{F} = m\vec{a}$ then fails, and one must return to the general statement

$$\sum \vec{F} = \frac{d(m\vec{v})}{dt}.$$

Applying this to a rocket ejecting gas at speed v_e relative to the rocket leads (the full derivation is beyond GCE, but the result is instructive) to the **rocket equation**:

$$m \frac{dv}{dt} = v_e \left(-\frac{dm}{dt} \right) + F_{\text{ext}},$$

where $-dm/dt > 0$ is the rate at which fuel mass is expelled. The thrust $v_e(-dm/dt)$ pushes the rocket forward even in the vacuum of space, where there is nothing to “push against” — a beautiful confirmation that momentum, not air, is what matters. For the GCE, you should simply know that the Second Law in momentum form handles such systems, while $F = ma$ does not.

KEY POINTS Key points of this section

- Momentum: $\vec{p} = m\vec{v}$; the Second Law states $\sum \vec{F} = d\vec{p}/dt$.
- Constant mass: $\sum \vec{F} = m\vec{a}$, applied in components after resolving all forces.
- Impulse: $\vec{J} = \int \vec{F} dt = \Delta\vec{p}$, the area under the F - t graph.
- Method: choose the system, draw the free-body diagram, apply $\sum \vec{F} = m\vec{a}$ per axis, then use kinematics.
- Classic models: inclined plane gives $a = g(\sin \theta - \mu_k \cos \theta)$; Atwood machine gives $a = \frac{(m_1 - m_2)g}{m_1 + m_2}$; braking distance scales as u^2 .
- Variable mass (rockets): use $\sum \vec{F} = d(m\vec{v})/dt$, never $m\vec{a}$ alone.

EXERCISE 2 Practice

1. A minibus of mass 1800 kg accelerates from rest to 15 m s^{-1} in 10 s. Find the net force acting on it, and the impulse delivered by the engine thrust (assumed to be the only horizontal force).

2. A block of mass 4.0 kg rests on a plane inclined at 30° to the horizontal. The coefficient of kinetic friction is 0.20. Find its acceleration down the plane. (Take $g = 9.81 \text{ m s}^{-2}$.)
3. In an Atwood machine, $m_1 = 5.0 \text{ kg}$ and $m_2 = 3.0 \text{ kg}$. Find the acceleration and the tension in the string.
4. A car travelling at 20 m s^{-1} brakes with a constant deceleration of 5.0 m s^{-2} . Find the braking distance, and comment on the effect of doubling the initial speed.

SOLUTION Exercise 1

1. $a = \frac{15 - 0}{10} = 1.5 \text{ m s}^{-2}$; $F = ma = 1800 \times 1.5 = 2700 \text{ N}$. Impulse $J = \Delta p = 1800 \times 15 = 27\,000 \text{ N s}$ (check: $J = F\Delta t = 2700 \times 10$).
2. $a = g(\sin 30^\circ - 0.20 \cos 30^\circ) = 9.81(0.5 - 0.20 \times 0.866) = 3.21 \text{ m s}^{-2}$.
3. $a = \frac{(5 - 3)}{5 + 3} \times 9.81 = 2.45 \text{ m s}^{-2}$; $T = \frac{2 \times 5 \times 3}{8} \times 9.81 = 36.8 \text{ N}$.
4. $s = \frac{u^2}{2a} = \frac{400}{10} = 40 \text{ m}$. Doubling u to 40 m s^{-1} gives $s = 160 \text{ m}$: four times farther.

DID YOU KNOW? Why seatbelts are Newton's Second Law in action

In a crash at 50 km h^{-1} , an unbelted passenger stops in roughly 0.01 s on the dashboard, while a belted passenger stops over about 0.15 s as the belt stretches. The change of momentum Δp is the same, but the average force $F = \Delta p / \Delta t$ is about fifteen times smaller with the belt. The Second Law, quietly, saves lives every day on the roads of Cameroon.

3 ACTION-REACTION AND THE THIRD LAW

In the previous section we learnt that a net external force acting on a body produces an acceleration in proportion to that force: $\vec{F} = m\vec{a}$. But where do forces come from in the first place? A force is never a property of a single isolated body; it is always the symptom of an interaction between two bodies. When you push a wheelbarrow, the wheelbarrow pushes back on your hands. When a pirogue's paddle pushes water backwards on the Wouri river, the water pushes the pirogue forwards. Newton's third law is the precise statement of this remarkable symmetry of nature, and it is the key to understanding propulsion, contact forces, and the behaviour of systems of connected bodies.

3.1 Observation: forces always come in pairs

EXPERIMENT A simple observation

Sit on a chair with wheels (or stand on a skateboard) facing a wall, and push the wall away from you. You and the chair roll backwards, away from the wall. Yet you pushed the wall forwards. The only way to explain your motion is that the wall exerted a force on you, directed away from it. Similarly, place two trolleys with spring-loaded bumpers in contact on a smooth bench and release them: both trolleys move apart, each accelerating away from the other. Neither trolley can push the other without being pushed itself.

The lesson of these observations is universal: whenever body A exerts a force on body B , body B simultaneously exerts a force on body A . There is no such thing as a one-sided force.

3.2 Statement of the Third Law

PROPERTY Newton's Third Law of Motion

If body A exerts a force $\vec{F}_{AB}$ on body B , then body B exerts a force $\vec{F}_{BA}$ on body A such that

$$\vec{F}_{BA} = -\vec{F}_{AB}.$$

The two forces are equal in magnitude, opposite in direction, and act along the same line. This law holds for all types of interaction (contact, gravitational, electric, magnetic) and whether the bodies are at rest or accelerating. (The statement itself is examinable; no formal proof is required, but you must be able to apply it rigorously.)

DEFINITION Action–reaction pair

An **action–reaction pair** is a pair of forces that:

1. are equal in magnitude;
2. are opposite in direction;
3. act on different bodies; and
4. arise from the same interaction, so they are of the same physical type (both gravitational, both contact, both frictional, etc.).

If one force of the pair is called the action, the other is the reaction; the choice of which is which is purely conventional.

WARNING The four characteristics are all essential

Two forces of equal magnitude and opposite direction are not necessarily an action–reaction pair. The decisive tests are: (i) do they act on different bodies? (ii) do they come from the same interaction? If either test fails, the forces are not a third-law pair.

3.3 Identifying action–reaction pairs**METHOD** Identifying action–reaction pairs

1. Choose one force acting in the situation, and write it with two subscripts: $\vec{F}_{\text{exerting, receiving}}$. For example $\vec{F}_{\text{Earth, book}}$ is the gravitational pull of the Earth on the book (the book's weight).
2. Identify the two bodies involved in that single interaction.
3. The reaction is the force exchanged in the same interaction, with the subscripts reversed: $\vec{F}_{\text{book, Earth}}$, the gravitational pull of the book on the Earth.
4. Check: the two forces act on different bodies, have the same magnitude, opposite directions, and the same physical nature.

EXAMPLE Everyday action–reaction pairs

- Book on a table. The table pushes the book upwards (normal contact force on the book); the book pushes the table downwards (normal contact force on the table). Same interaction, different bodies.
- Your weight. The Earth pulls you downwards with force mg ; you pull the Earth upwards with a force of magnitude mg . Because the Earth's mass is enormous, its acceleration is utterly negligible.
- Walking. Your foot pushes the ground backwards and downwards; the ground pushes your foot forwards and upwards. It is the ground's forward push on you that accelerates you along the road.

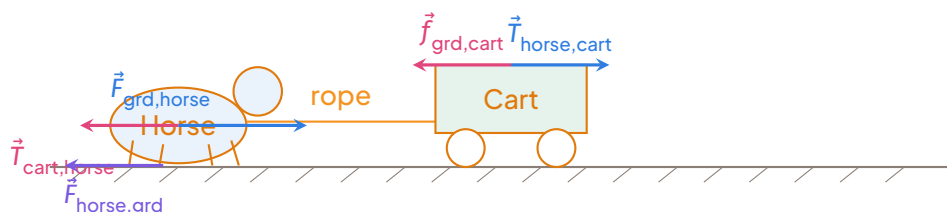
WARNING Weight and normal reaction are NOT an action–reaction pair

This is the most frequent error at the GCE. Consider a book resting on a table. Its weight $\vec{W}$ (Earth pulls book, downwards) and the normal reaction $\vec{R}$ of the table (table pushes book,

upwards) are equal and opposite when the book is in equilibrium — but they both act on the same body (the book) and they arise from different interactions (gravitational for $\vec{W}$, contact for $\vec{R}$). Proof that they are not a pair: if you suddenly lift the table, $\vec{R}$ changes instantly while $\vec{W}$ does not; and in a lift accelerating upwards, $R > mg$, so they are not even equal. The true reaction to $\vec{W}$ is the book's gravitational pull on the Earth; the true reaction to $\vec{R}$ is the book's downward push on the table.

3.4 Free-body diagrams for interacting bodies

To apply the second law correctly to each body of a system, we draw a separate **free-body diagram** for each body, showing only the forces acting on that body. The third law then tells us how a force appearing in one diagram is related to a force in the other.



Horse and cart. The horse pushes the ground backwards ($\vec{F}_{\text{horse,grd}}$); the ground pushes the horse forwards ($\vec{F}_{\text{grd,horse}}$): a third-law pair. The rope tensions $\vec{T}_{\text{cart,horse}}$ and $\vec{T}_{\text{horse,cart}}$ form another pair.

EXAMPLE The horse–cart paradox

A student objects: “If the cart pulls the horse backwards with a force equal and opposite to the force with which the horse pulls the cart forwards, the two forces cancel and the system can never move!” Resolve the paradox.

Resolution. The two forces of an action–reaction pair act on different bodies, so they can never cancel each other. $\vec{T}_{\text{horse,cart}}$ acts on the cart alone; $\vec{T}_{\text{cart,horse}}$ acts on the horse alone. Whether the horse accelerates forwards depends on the forces on the horse: the ground pushes it forwards with $\vec{F}_{\text{grd,horse}}$ while the rope pulls it backwards with $\vec{T}_{\text{cart,horse}}$. The horse–cart system accelerates because the ground exerts a net forward external force on it. The horse moves forward because it pushes the ground backwards, not merely because it pulls the cart.

3.5 Propulsion: the rocket and the swimmer

Propulsion is the most spectacular illustration of the third law. A rocket does not “push against the air”; it works perfectly in the vacuum of space. The engine expels hot gas backwards at high speed: the rocket exerts a backward force on the gas, and the gas exerts an equal forward force — the **thrust** — on the rocket.